# Voice Assistance Proposal

## I. Introduction

Voice Assistance is a Python text-to-speech (TTS) system that can transfer from text to voice. We are going to build a desktop mini application. Which can overlay on Microsoft Word and other apps for ease of use. We work on text and sound related to the Data Science field, which is a first step toward deep learning in AI and machine learning.

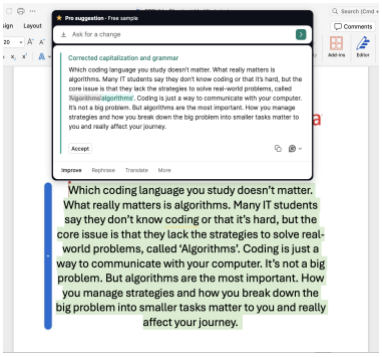
## II. Objective & Proposed Solution

Many Cambodians lack reading skills, and they feel exhausted from reading, especially for students, who have too many documents to read every day. To improve their learning journey, we have proposed a solution with the technology TTS-system using the pyttsx3 library. It is a lab study tool that can change student behavior from being lazy in reading to being ready to listen.

Future features

| **Core Feature** | **Description** | **Library & PyConcept** |
|---|---|---|
| TTS-Engine (core) | It is core functionality in our system that can convert from text to speech. | file & error handling, OOP, pyttsx3, playsound, keyboard |
| Overlay and Text-Select-Tricker | It is the enhance feature of our system. Make our mini-lab-app stand out and useful. | Pillow, keyboard opencv-python |
| GUI (Mostly) | Just some tab of GUI for ease of use. And use QT-framework | QtWidgets, Qthread, QSS, Signals & Slots, event-handling |

Just select, and our tool can know how much you want it to read for you.
Ex. (Overlay on Word like Grammarly)




How we apply OOP: Create a class TTSEngine, and apply all concepts of OOP (To extend the class and hide the complex logic)
File handling: We create a temp-file to store the text from user input and clean it.
Error handling: To avoid any errors, we use a try block when passing the text to the function generate speech and an except to show an error message.

## III. Methodology

Step1: Research each task that have assigned. To build fundamental

Member 1: research how the TTS system works and library pyttsx3
Member 2: How can our system overlay on others app and text select tracking workflow?
Member3: Design GUI, learn basic PyQt6, QSS, Q-Thread

Step2: Create Project Structure --- Git ----

Step3: Start implementing cod

Member1: Implement offline TTS (pyttsx3) apply OOP (TTS Engine class)
Member2: Implement global hotkey, Capture selected text via clipboard
Member3: Implement basic GUI

Step4: Binding code and connecting to GUI

| **Tool** | **Library** | **Framework** |
|---|---|---|
| **VS Code**<br>**QT 6.10.2**<br>**Git** | pyttsx3, playsound, keyboard<br>pillow, keyboard, opencv-python<br>QtWidgets, QSS, QThread | **PyQt6** |

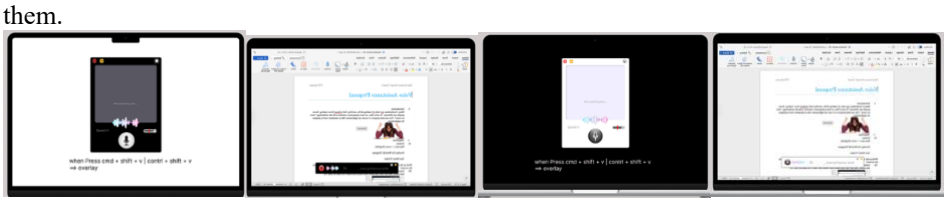
Case1: If the user selects more than 250 words. Warning!! You can only read 250 words at a time.
Case2: If the user presses random text. Warning!! Are you sure that is actual English words?
Case3: …

## IV. Development Timeline (3 Weeks)

| Team | Role | W1:27th | W2 | W3 | W4 |
|---|---|---|---|---|---|
| Keo Seavpav | Core TTS system | Concept understand | Build TTS Engine & apply OOP | Update progress | combine |
| Hong Limhak | Overlay & text select tricker | Concept understand | Support Team | Update progress | combine |
| Vean Sovanvirak | GUI integrations | Design Fixma Explore PyQT6 | Start implement code | Update progress | combine |

## V. Expected Outcome

Flow GUI
After the user selects text and press speak button, our system being process generate voice and speaks to them.



Expectation
- Understand the technologies
- Apply the concept of OOP
- Learn about the TTS system
- Python library

## VI. References

https://doc.qt.io
https://www.riverbankcomputing.com/static/Docs/PyQt6/
https://pypi.org/
https://pyttsx3.readthedocs.io/en/latest/engine.html

Note: We work on English language only.